# ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** tiny particles | Matter consists of very small particles. |
| 2 | **(c)** melting | Fusion. |
| 3 | **(d)** increase in temperature | Warmer liquids evaporate faster. |
| 4 | **(d)** sugar dissolves in water without raising the water level much | Sugar particles occupy spaces between water particles. |
| 5 | **(c)** evaporation | Surface process. |
| 6 | **(d)** spread out in a windy place | More surface and wind. |
| 7 | **(b)** constantly moving | Particles are always in motion. |
| 8 | **(c)** melting point | Melting point. |
| 9 | **(c)** cooling | Fastest particles leave, so cooling. |
| 10 | **(c)** diffusion | Particles of ink mix with water particles. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Matter is made of tiny particles (1); Particles move and have spaces between them (1). | 1+1 |
| 12 | Evaporation occurs at any temperature; boiling at a fixed temperature (1); Evaporation is at the surface; boiling throughout the liquid (1). | 1+1 |
| 13 | Condensation (1); Water vapour in air cools and becomes liquid (1). | 1+1 |
| 14 | Spontaneous mixing of particles of two substances (1); Smell of perfume spreading in a room (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Dissolve a spoon of salt/sugar in a beaker of water (1); Level of water hardly rises (1); Solute particles fit in the gaps between water particles (1). | 1+1+1 |
| 16 | Fastest particles escape from the liquid (1); Remaining particles have lower energy (1); Temperature of the liquid falls (1). | 1+1+1 |
| 17 | Water vapour in air touches the cold glass (1); It loses energy and condenses (1); Droplets form (1). | 1+1+1 |
| | **OR:** Put a crystal of potassium permanganate in water (1); Colour spreads slowly (1); Particles move and mix on their own (1). | 1+1+1 |
| 18 | Cool it (1); Compress it (1); Particles come closer and attract each other (1). | 1+1+1 |
| | **OR:** Temperature rises when heating (1); Flat at 0: melting (1); Flat at 100: boiling (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Dissolving sugar (1); Spreading of ink (1); Smell of perfume (1); Spaces between particles (1); Particles in constant motion (1). | 1+1+1+1+1 |
| | **OR:** Wrap cotton on the bulb (1); Dip in water, then in acetone (1); Note the temperature fall (1); Acetone gives more cooling (1); Faster evaporation leads to more cooling (1). | 1+1+1+1+1 |
| 20 | Shape (1); Volume (1); Compressibility (1); Particle arrangement (1); Forces between particles (1). | 1+1+1+1+1 |
| | **OR:** Particles move randomly (1); They collide with container walls (1); Collisions exert force (1); More collisions give more pressure (1); Heating increases the pressure (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Saying that particles of a solid are at rest; they vibrate about fixed positions.
- Mixing up the spacing: gas particles are far apart, liquid and solid particles are close.
- Saying temperature keeps rising during melting or boiling; it stays constant during a change of state.
- Writing that evaporation happens only at the boiling point, or that it heats the surroundings; it causes cooling.

- Confusing diffusion (mixing of particles by themselves) with dissolving or with stirring.
- Stating that a liquid has a definite shape; it has a definite volume only.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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# ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 0 degrees Celsius | Ice melts at 0. |
| 2 | **(c)** cools the body | Cooling by evaporation. |
| 3 | **(d)** gases | Gas particles move quickly with big gaps. |
| 4 | **(b)** 100 degrees Celsius | Water boils at 100. |
| 5 | **(b)** amount of water vapour in the air | Moisture content. |
| 6 | **(c)** solids | Strong forces hold them in place. |
| 7 | **(c)** condensation | Gas to liquid. |
| 8 | **(d)** decreases | Air is already saturated. |
| 9 | **(d)** liquid | Liquids take the shape of the container. |
| 10 | **(d)** sublimation | Camphor and naphthalene. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | It evaporates quickly (1); Takes heat from the skin (1). | 1+1 |
| 12 | Particles move faster at higher temperature (1); Diffusion is faster (1). | 1+1 |
| 13 | Particles move rapidly in all directions (1); Forces between them are very weak (1). | 1+1 |
| 14 | Direct change of a solid into a gas (1); Camphor, naphthalene, dry ice (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Solids: close, fixed, vibrate (1); Liquids: close, can slide past (1); Gases: far apart, move freely (1). | 1+1+1 |
| 16 | Melting ice absorbs a large amount of heat from the drink (1); More heat is taken from the drink (1); Water at 0 degrees Celsius absorbs heat only by warming up (1). | 1+1+1 |
| 17 | Camphor, naphthalene, dry ice (1); Dew, mist, fog (1); Water droplets on glass (1). | 1+1+1 |
|  | **OR:** Solid: fixed shape (1); Liquid: fixed volume, no fixed shape (1); Gas: no fixed shape or volume (1). | 1+1+1 |
| 18 | Steam carries extra heat that was absorbed during vaporisation (1); It gives this heat out on condensing on the skin (1); More heat is transferred to the skin (1). | 1+1+1 |
|  | **OR:** Particles move faster when hot (1); Faster diffusion (1); Hot vapour spreads quickly (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Melting (1); Freezing (1); Evaporation/boiling (1); Condensation (1); Sublimation (1). | 1+1+1+1+1 |
|  | **OR:** Strong forces in solids (1); Weaker in liquids (1); Very weak in gases (1); Heating weakens the forces (1); Cooling strengthens them (1). | 1+1+1+1+1 |
| 20 | Surface process at any temperature (1); Temperature, surface area, wind, humidity (1); Drying clothes (1); Cooling of earthen pot (1); Sweating (1). | 1+1+1+1+1 |
|  | **OR:** Same cloth and water (1); Vary one factor only (1); Measure drying time (1); Higher temperature and wind dry faster (1); Humidity slows it (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Saying that particles of a solid are at rest; they vibrate about fixed positions.
- Mixing up the spacing: gas particles are far apart, liquid and solid particles are close.
- Saying temperature keeps rising during melting or boiling; it stays constant during a change of state.
- Writing that evaporation happens only at the boiling point, or that it heats the surroundings; it causes cooling.
- Confusing diffusion (mixing of particles by themselves) with dissolving or with stirring.
- Stating that a liquid has a definite shape; it has a definite volume only.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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# ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** water evaporates through tiny pores of the pot | Evaporation cools. |
| 2 | **(c)** gas | Gas fills the whole container. |
| 3 | **(c)** camphor | Camphor changes directly. |
| 4 | **(d)** increasing pressure and lowering temperature | Compress and cool. |
| 5 | **(d)** gases | Large gaps between particles. |
| 6 | **(c)** remains constant | Energy used to change state. |
| 7 | **(b)** liquid under high pressure | Gas liquefied by pressure. |
| 8 | **(d)** solids | Particles are tightly packed. |
| 9 | **(b)** overcome the forces of attraction between particles | The temperature stays constant during the change of state. |
| 10 | **(a)** carbon dioxide | It sublimes. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Large spaces between particles (1); Particles can be pushed closer (1). | 1+1 |
| 12 | Particles are tightly packed (1); Very little space is available (1). | 1+1 |
| 13 | Heat is used to break forces between particles (1); Not to raise the temperature (1). | 1+1 |
| 14 | Gas is compressed under high pressure (1); It turns into a liquid (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Temperature (1); Surface area and wind speed (1); Humidity (1). | 1+1+1 |
| 16 | Smell of cooking (1); Perfume spreading (1); Tea dispersing in water (1). | 1+1+1 |
| 17 | Particles gain energy and move faster (1); At boiling point they break free throughout (1); Bubbles of vapour form (1). | 1+1+1 |
| | **OR:** Low humidity speeds evaporation (1); More cooling is achieved (1); In humid weather it is less effective (1). | 1+1+1 |
| 18 | Melting point: solid to liquid (1); Boiling point: liquid to gas (1); Both are fixed for a pure substance (1). | 1+1+1 |
| | **OR:** Compressibility: solids none, liquids little, gases high (1); Density: highest in solids typically (1); Shape: solids fixed (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Heat energy supplied is used to overcome forces between particles (1); It is not used to increase the motion of particles (1); So the temperature stays constant during melting (1); and also during boiling (1); Temperature rises again after the change of state is complete (1). | 1+1+1+1+1 |
| | **OR:** Camphor changes to vapour (1); Salt does not (1); Heat mixture in a china dish with an inverted funnel (1); Camphor collects on funnel (1); Pure salt remains (1). | 1+1+1+1+1 |
| 20 | Cooling lowers particle energy (1); Pressure pushes particles closer (1); LPG is liquefied under pressure (1); Dry ice sublimes at normal pressure (1); Gases can be liquefied (1). | 1+1+1+1+1 |
| | **OR:** Hot vapour near the spout is invisible (1); Cooler air makes it condense (1); Tiny droplets form (1); Droplets scatter light as a white cloud (1); This is condensation (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

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- Mixing up the spacing: gas particles are far apart, liquid and solid particles are close.
- Saying temperature keeps rising during melting or boiling; it stays constant during a change of state.
- Writing that evaporation happens only at the boiling point, or that it heats the surroundings; it causes cooling.
- Confusing diffusion (mixing of particles by themselves) with dissolving or with stirring.

- Stating that a liquid has a definite shape; it has a definite volume only.

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